

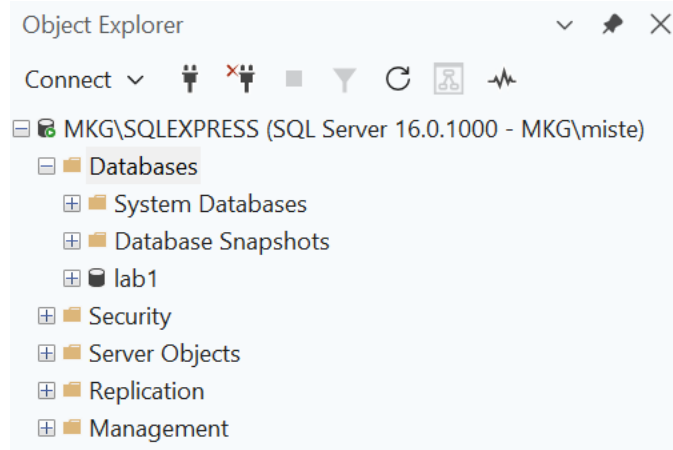
DATABASE: lab1

1. Create a database lab1

SQL Query:

```
CREATE DATABASE lab1;
```

Output:

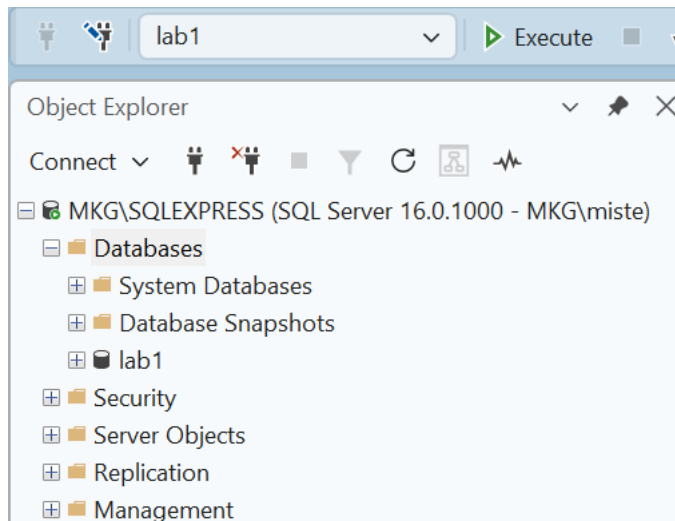


2. Connect database.

SQL Query:

```
USE lab1;
```

Output:

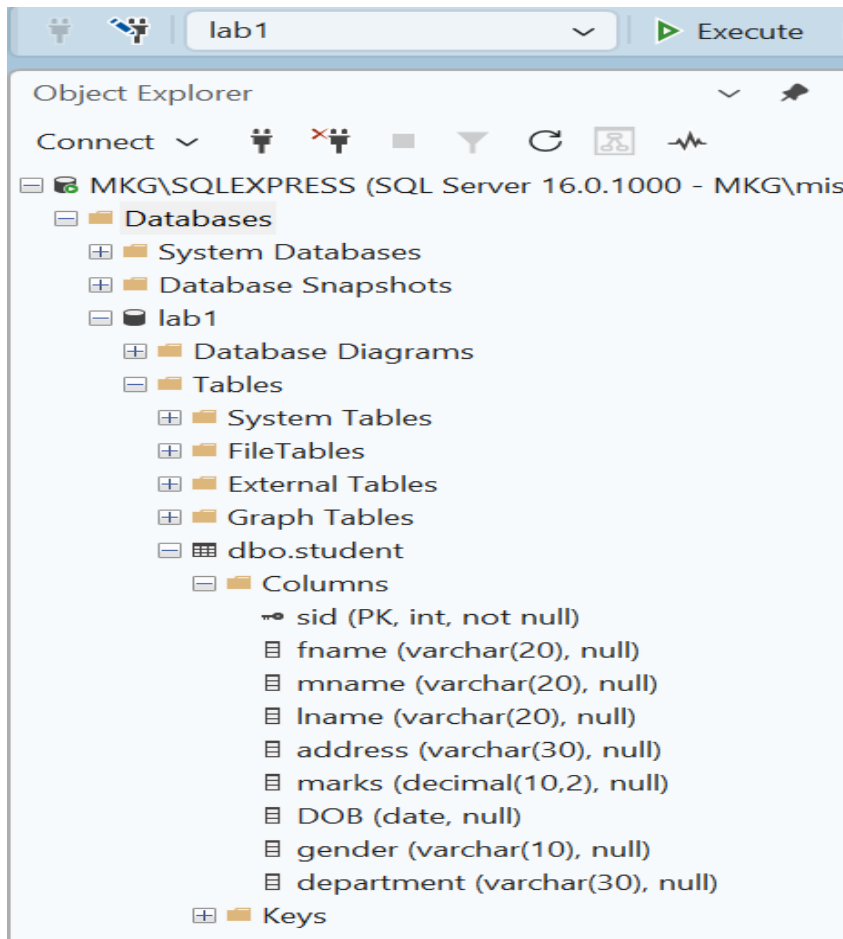


3. Create a student table (sid, fname , mname, lname, address, marks, dob, gender, department)

SQL Query:

```
CREATE TABLE student(  
    sid int primary key,  
    fname varchar (20),  
    mname varchar (20),  
    lname varchar (20),  
    address varchar (30),  
    marks decimal (10,2),  
    DOB date,  
    gender varchar (10),  
    department varchar (30)  
);
```

Output:



4. Insert any 20 records to given table.

SQL Query:

INSERT INTO student values

```
(1,'Ram', 'Prasad','Bhattarai','Gongabu',130.00,'2062-07-26','Male','IT'),
(2,'Sam',", 'Adhikari','Kadaghari',199.00,'2063-07-21','Male','IT'),
(3,'Sita',", 'Bhatt','Tinkune',199.00,'2064-04-10','Female','IT'),
(4,'Hari',", 'Acharya','Kalanki',170.00,'2062-01-08','Male','IT'),
(5,'Pratik',", 'Acharya', 'Central Jail',180.00,'2062-03-13','Male','IT'),
(6,'Sabal',", 'Thapa','Kumarigal',175.25,'2062-10-21','Male','IT'),
(7,'Pupen','Hang','Limbu',Shantinagar ',174.50,'2063-04-15','Male','IT'),
(8,'Sujan',", 'Adhikari','Tokha',180.00,'2062-04-17','Male','IT'),
(9,'Chandra','Dev','Yadav','Balkumari',160.00,'2058-09-03','Male','IT'),
(10,'Prawjol',", 'Gurung','Bhaktapur',129.50,'2062-12-31','Male','IT'),
(11,'Karun',", 'Gyawali','Shantinagar',190.00,'2063-08-08','Male','IT'),
(12,'Rabi',", 'Bhatt','Shantinagar',150.00,'2063-10-02','Male','IT'),
(13,'Sandesh','Bandu','Shrestha','Baneshwor',160.00,'2063-05-15','Male','IT'),
(14,'Prasis',", 'Karki','Suncity',170.00,'2063-02-19','Male','IT'),
(15,'Roshan','Kumar','Thakur','Shantinagar',200.00,'2063-12-02','Male','IT'),
(16,'Dhirga','Narayan','Shrestha','Bhaktapur',198.00,'2052-05-25','Male','IT'),
(17,'Shital',", 'Bhusal','Sankhamul',145.00,'2062-11-02','Male','IT'),
(18,'Samrat',", 'Paudel','Sankhamul',185.00,'2062-08-24','Male','IT'),
(19,'Prajwol',", 'Ghorasaine','Koteswor',160.00,'2063-06-20','Male','IT'),
(20,'Saras',", 'Ghatani','Dhumbarai',45.00,'2064-06-12','Male','IT');
SELECT *FROM student;
```

Output:

Results Messages									
	sid	fname	mname	lname	address	marks	DOB	gender	department
1	1	Ram	Prasad	Bhattarai	Gongabu	130.00	2062-07-26	Male	IT
2	2	Sam		Adhikari	Kadaghari	199.00	2063-07-21	Male	IT
3	3	Sita		Bhatt	Tinkune	199.00	2064-04-10	Female	IT
4	4	Hari		Acharya	Kalanki	170.00	2062-01-08	Male	IT
5	5	Pratik		Acharya	Central Jail	180.00	2062-03-13	Male	IT
6	6	Sabal		Thapa	Kumarigal	175.25	2062-10-21	Male	IT
7	7	Pupen	Hang	Limbu	Shantinagar	174.50	2063-04-15	Male	IT
8	8	Sujan		Adhikari	Tokha	180.00	2062-04-17	Male	IT
9	9	Chandra	Dev	Yadav	Balkumari	160.00	2058-09-03	Male	IT
10	10	Prawjol		Gurung	Bhaktapur	129.50	2062-12-31	Male	IT
11	11	Karun		Gyawali	Shantinagar	190.00	2063-08-08	Male	IT
12	12	Rabi		Bhatt	Shantinagar	150.00	2063-10-02	Male	IT
13	13	Sande...	Bandu	Shrestha	Baneshwor	160.00	2063-05-15	Male	IT
14	14	Prasis		Karki	Suncity	170.00	2063-02-19	Male	IT
15	15	Roshan	Kumar	Thakur	Shantinagar	200.00	2063-12-02	Male	IT
16	16	Dhirga	Naray...	Shrestha	Bhaktapur	198.00	2052-05-25	Male	IT
17	17	Shital		Bhusal	Sankhamul	145.00	2062-11-02	Male	IT
18	18	Samrat		Paudel	Sankhamul	185.00	2062-08-24	Male	IT
19	19	Prajwol		Ghoras...	Koteswor	160.00	2063-06-20	Male	IT
20	20	Saras		Ghatani	Dhumbarai	45.00	2064-06-12	Male	IT

5. Display only name and marks with name in single column.

SQL Query:

```
SELECT CONCAT (fname,' ',COALESCE(mname,"'),' ',lname) as Name, marks  
FROM student;
```

OUTPUT:

	Name	marks
1	Ram Prasad Bhattarai	130.00
2	Sam Adhikari	199.00
3	Sita Bhatt	199.00
4	Hari Acharya	170.00
5	Pratik Acharya	180.00
6	Sabal Thapa	175.25
7	Pupen Hang Limbu	174.50
8	Sujan Adhikari	180.00
9	Chandra Dev Yadav	160.00
10	Prawjol Gurung	129.50
11	Karun Gyawali	190.00
12	Rabi Bhatt	150.00
13	Sandesh Bandu Shrestha	160.00
14	Prasis Karki	170.00
15	Roshan Kumar Thakur	200.00
16	Dhirga Narayan Shrestha	198.00
17	Shital Bhusal	145.00
18	Samrat Paudel	185.00
19	Prajwol Ghorasaine	160.00
20	Saras Ghatani	45.00

6. Find name of those student who are from 'Shantinagar' and get marks greater than 75.

SQL Query:

```
SELECT CONCAT (fname,' ',COALESCE(mname,"'),' ',lname) AS Name, marks, address  
FROM student  
WHERE address = 'Shantinagar' AND marks > '100';
```

OUTPUT:

	Name	marks	address
1	Pupen Hang Limbu	174.50	Shantinagar
2	Karun Gyawali	190.00	Shantinagar
3	Rabi Bhatt	150.00	Shantinagar
4	Roshan Kumar Thakur	200.00	Shantinagar

7. Change address of all students to 'Palpa' who get marks less than 100.

SQL Query:

SELECT *

FROM student;

UPDATE student

SET address = 'Palpa'

WHERE marks < 100;

OUTPUT:

Results Messages									
	sid	fname	mname	lname	address	marks	DOB	gender	department
1	1	Ram	Prasad	Bhattarai	Gongabu	130.00	2062-07-26	Male	IT
2	2	Sam		Adhikari	Kadaghari	199.00	2063-07-21	Male	IT
3	3	Sita		Bhatt	Tinkune	199.00	2064-04-10	Female	IT
4	4	Hari		Acharya	Kalanki	170.00	2062-01-08	Male	IT
5	5	Pratik		Acharya	Central Jail	180.00	2062-03-13	Male	IT
6	6	Sabal		Thapa	Kumarigal	175.25	2062-10-21	Male	IT
7	7	Pupen	Hang	Limbu	Shantinagar	174.50	2063-04-15	Male	IT
8	8	Sujan		Adhikari	Tokha	180.00	2062-04-17	Male	IT
9	9	Chandra	Dev	Yadav	Balkumari	160.00	2058-09-03	Male	IT
10	10	Prawjol		Gurung	Bhaktapur	129.50	2062-12-31	Male	IT
11	11	Karun		Gyawali	Shantinagar	190.00	2063-08-08	Male	IT
12	12	Rabi		Bhatt	Shantinagar	150.00	2063-10-02	Male	IT
13	13	Sandesh	Bandu	Shrestha	Baneshwor	160.00	2063-05-15	Male	IT
14	14	Prasis		Karki	Suncity	170.00	2063-02-19	Male	IT
15	15	Roshan	Kumar	Thakur	Shantinagar	200.00	2063-12-02	Male	IT
16	16	Dhirga	Narayan	Shrestha	Bhaktapur	198.00	2052-05-25	Male	IT
17	17	Shital		Bhusal	Sankhamul	145.00	2062-11-02	Male	IT
18	18	Samrat		Paudel	Sankhamul	185.00	2062-08-24	Male	IT
19	19	Prajwol		Ghorasaine	Koteshwor	160.00	2063-06-20	Male	IT
20	20	Saras		Ghatani	Dhumbarai	45.00	2064-06-12	Male	IT

8. Find all students whose address start with 'k' or end with 'r'

SQL Query:

SELECT *

FROM student

WHERE address LIKE 'k%'

OR address LIKE '%r';

OUTPUT:

Results Messages									
	sid	fname	mname	lname	address	marks	DOB	gender	department
1	2	Sam		Adhikari	Kadaghari	199.00	2063-07-21	Male	IT
2	4	Hari		Acharya	Kalanki	170.00	2062-01-08	Male	IT
3	6	Sabal		Thapa	Kumarigal	175.25	2062-10-21	Male	IT
4	7	Pupen	Hang	Limbu	Shantinagar	174.50	2063-04-15	Male	IT
5	10	Prawjol		Gurung	Bhaktapur	129.50	2062-12-31	Male	IT
6	11	Karun		Gyawali	Shantinagar	190.00	2063-08-08	Male	IT
7	12	Rabi		Bhatt	Shantinagar	150.00	2063-10-02	Male	IT
8	13	Sandesh	Bandu	Shrestha	Baneshwor	160.00	2063-05-15	Male	IT
9	15	Roshan	Kumar	Thakur	Shantinagar	200.00	2063-12-02	Male	IT
10	16	Dhirga	Narayan	Shrestha	Bhaktapur	198.00	2052-05-25	Male	IT
11	19	Prajwol		Ghorasaine	Koteshwor	160.00	2063-06-20	Male	IT

9. Add a new field phone number of datatype varchar(10) and insert 20 phone number to given student table.

SQL Query:

ALTER TABLE student

ADD PhoneNo VARCHAR(10);

```
UPDATE student SET PhoneNo = '9844000001' WHERE sid = 1;
UPDATE student SET PhoneNo = '9844000002' WHERE sid = 2;
UPDATE student SET PhoneNo = '9844000003' WHERE sid = 3;
UPDATE student SET PhoneNo = '9844000004' WHERE sid = 4;
UPDATE student SET PhoneNo = '9844000005' WHERE sid = 5;
UPDATE student SET PhoneNo = '9844000006' WHERE sid = 6;
UPDATE student SET PhoneNo = '9844000007' WHERE sid = 7;
UPDATE student SET PhoneNo = '9844000008' WHERE sid = 8;
UPDATE student SET PhoneNo = '9844000009' WHERE sid = 9;
UPDATE student SET PhoneNo = '9844000010' WHERE sid = 10;
UPDATE student SET PhoneNo = '9844000011' WHERE sid = 11;
UPDATE student SET PhoneNo = '9844000012' WHERE sid = 12;
UPDATE student SET PhoneNo = '9844000013' WHERE sid = 13;
UPDATE student SET PhoneNo = '9844000014' WHERE sid = 14;
UPDATE student SET PhoneNo = '9844000015' WHERE sid = 15;
UPDATE student SET PhoneNo = '9844000016' WHERE sid = 16;
UPDATE student SET PhoneNo = '9844000017' WHERE sid = 17;
UPDATE student SET PhoneNo = '9844000018' WHERE sid = 18;
UPDATE student SET PhoneNo = '9844000019' WHERE sid = 19;
UPDATE student SET PhoneNo = '9844000020' WHERE sid = 20;
SELECT *FROM student;
```

OUTPUT:

Results Messages										
	sid	fname	mname	lname	address	marks	DOB	gender	department	PhoneNo
1	1	Ram	Prasad	Bhattarai	Gongabu	130.00	2062-07-26	Male	IT	9844000001
2	2	Sam		Adhikari	Kadaghari	199.00	2063-07-21	Male	IT	9844000002
3	3	Sita		Bhatt	Tinkune	199.00	2064-04-10	Female	IT	9844000003
4	4	Hari		Acharya	Kalanki	170.00	2062-01-08	Male	IT	9844000004
5	5	Pratik		Acharya	Central Jail	180.00	2062-03-13	Male	IT	9844000005
6	6	Sabal		Thapa	Kumarigal	175.25	2062-10-21	Male	IT	9844000006
7	7	Pupen	Hang	Limbu	Shantinagar	174.50	2063-04-15	Male	IT	9844000007
8	8	Sujan		Adhikari	Tokha	180.00	2062-04-17	Male	IT	9844000008
9	9	Chandra	Dev	Yadav	Balkumari	160.00	2058-09-03	Male	IT	9844000009
10	10	Prawjol		Gurung	Bhaktapur	129.50	2062-12-31	Male	IT	9844000010
11	11	Karun		Gyawali	Shantinagar	190.00	2063-08-08	Male	IT	9844000011
12	12	Rabi		Bhatt	Shantinagar	150.00	2063-10-02	Male	IT	9844000012
13	13	Sandesh	Bandu	Shrestha	Baneshwor	160.00	2063-05-15	Male	IT	9844000013
14	14	Prasis		Karki	Suncity	170.00	2063-02-19	Male	IT	9844000014
15	15	Roshan	Kumar	Thakur	Shantinagar	200.00	2063-12-02	Male	IT	9844000015
16	16	Dhirga	Narayan	Shrestha	Bhaktapur	198.00	2052-05-25	Male	IT	9844000016
17	17	Shital		Bhusal	Sankhamul	145.00	2062-11-02	Male	IT	9844000017
18	18	Samrat		Paudel	Sankhamul	185.00	2062-08-24	Male	IT	9844000018
19	19	Prajwol		Ghorasaine	Koteshwor	160.00	2063-06-20	Male	IT	9844000019
20	20	Saras		Ghatani	Palpa	45.00	2064-06-12	Male	IT	9844000020

10. Find highest and lowest marks.

SQL Query:

```
SELECT MAX(marks) AS highest_marks,  
       MIN(marks) AS lowest_marks  
FROM student;
```

OUTPUT:

Results		Messages	
	highest_marks	lowest_marks	
1	200.00	45.00	

11. Display name of all students in alphabetically.

SQL Query:

```
SELECT CONCAT (fname,' ',COALESCE(mname,"),' ',lname) as Name  
FROM student  
ORDER BY Name ASC
```

OUTPUT:

Results		Messages	
	Name		
1	Chandra Dev Yadav		
2	Dhirga Narayan Shrestha		
3	Hari Acharya		
4	Karun Gyawali		
5	Prajwol Ghorasaine		
6	Prasis Karki		
7	Pratik Acharya		
8	Prawjol Gurung		
9	Pupen Hang Limbu		
10	Rabi Bhatt		
11	Ram Prasad Bhattarai		
12	Roshan Kumar Thakur		
13	Sabal Thapa		
14	Sam Adhikari		
15	Samrat Paudel		
16	Sandesh Bandu Shrestha		
17	Saras Ghatani		
18	Shital Bhusal		
19	Sita Bhatt		
20	Sujan Adhikari		

12. Find department having more than one student.

SQL Query:

```
SELECT department, COUNT(*) AS total_students
FROM student
GROUP BY department
HAVING COUNT(*) > 1;
```

OUTPUT:

Results Messages		
	department	total_students
1	IT	20

13. Delete all student who get marks less than 50

SQL Query:

```
DELETE
FROM student
WHERE marks<50
SELECT *
FROM student;
```

OUTPUT:

Results Messages										
	sid	fname	mname	lname	address	marks	DOB	gender	department	PhoneNo
1	1	Ram	Prasad	Bhattarai	Gongabu	130.00	2062-07-26	Male	IT	9844000001
2	2	Sam		Adhikari	Kadaghari	199.00	2063-07-21	Male	IT	9844000002
3	3	Sita		Bhatt	Tinkune	199.00	2064-04-10	Female	IT	9844000003
4	4	Hari		Acharya	Kalanki	170.00	2062-01-08	Male	IT	9844000004
5	5	Pratik		Acharya	Central Jail	180.00	2062-03-13	Male	IT	9844000005
6	6	Sabal		Thapa	Kumarigal	175.25	2062-10-21	Male	IT	9844000006
7	7	Pupen	Hang	Limbu	Shantinagar	174.50	2063-04-15	Male	IT	9844000007
8	8	Sujan		Adhikari	Tokha	180.00	2062-04-17	Male	IT	9844000008
9	9	Chandra	Dev	Yadav	Balkumari	160.00	2058-09-03	Male	IT	9844000009
10	10	Prawjol		Gurung	Bhaktapur	129.50	2062-12-31	Male	IT	9844000010
11	11	Karun		Gyawali	Shantinagar	190.00	2063-08-08	Male	IT	9844000011
12	12	Rabi		Bhatt	Shantinagar	150.00	2063-10-02	Male	IT	9844000012
13	13	Sandesh	Bandu	Shrestha	Baneshwor	160.00	2063-05-15	Male	IT	9844000013
14	14	Prasis		Karki	Suncity	170.00	2063-02-19	Male	IT	9844000014
15	15	Roshan	Kumar	Thakur	Shantinagar	200.00	2063-12-02	Male	IT	9844000015
16	16	Dhirga	Narayan	Shrestha	Bhaktapur	198.00	2052-05-25	Male	IT	9844000016
17	17	Shital		Bhusal	Sankhamul	145.00	2062-11-02	Male	IT	9844000017
18	18	Samrat		Paudel	Sankhamul	185.00	2062-08-24	Male	IT	9844000018
19	19	Prajwol		Ghorasaine	Koteshwor	160.00	2063-06-20	Male	IT	9844000019

```
SQL Query:
SELECT *
FROM student
WHERE DOB BETWEEN '2062-01-01' AND '2063-01-01';
```

Results		Messages								
	sid	fname	mname	lname	address	marks	DOB	gender	department	PhoneNo
1	1	Ram	Prasad	Bhattarai	Gongabu	130.00	2062-07-26	Male	IT	9844000001
2	4	Hari		Acharya	Kalanki	170.00	2062-01-08	Male	IT	9844000004
3	5	Pratik		Acharya	Central Jail	180.00	2062-03-13	Male	IT	9844000005
4	6	Sabal		Thapa	Kumarigal	175.25	2062-10-21	Male	IT	9844000006
5	8	Sujan		Adhikari	Tokha	180.00	2062-04-17	Male	IT	9844000008
6	10	Prawjol		Gurung	Bhaktapur	129.50	2062-12-31	Male	IT	9844000010
7	17	Shital		Bhusal	Sankhamul	145.00	2062-11-02	Male	IT	9844000017
8	18	Samrat		Paudel	Sankhamul	185.00	2062-08-24	Male	IT	9844000018

```
SQL Query:
SELECT COUNT(*) AS Total
FROM student
WHERE address = 'Palpa';
```

Results		Messages	
	Total		
1	0		

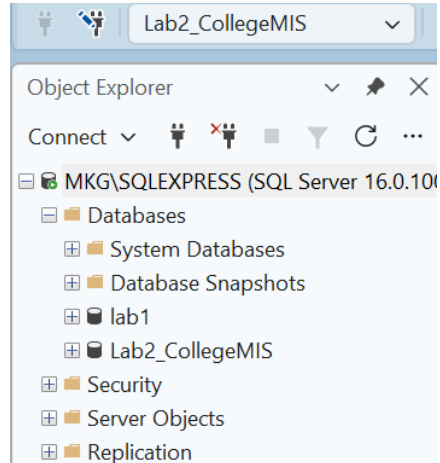
DATABASE: Lab2-CollegeMIS

1. Create database lab2_collegeMIS and connect database.

SQL Query:

```
CREATE DATABASE Lab2_CollegeMIS;  
USE Lab2_CollegeMIS;
```

OUTPUT:



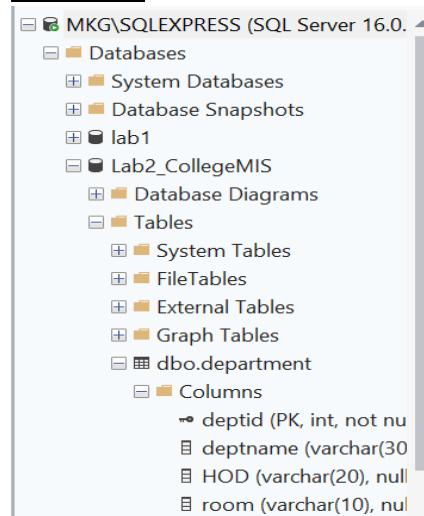
2. Create all 5 tables in order.

- a. Department:

SQL Query:

```
CREATE TABLE department(  
deptid INT PRIMARY KEY,  
deptname VARCHAR(30),  
HOD VARCHAR(20),  
room VARCHAR(10)  
);
```

OUTPUT:

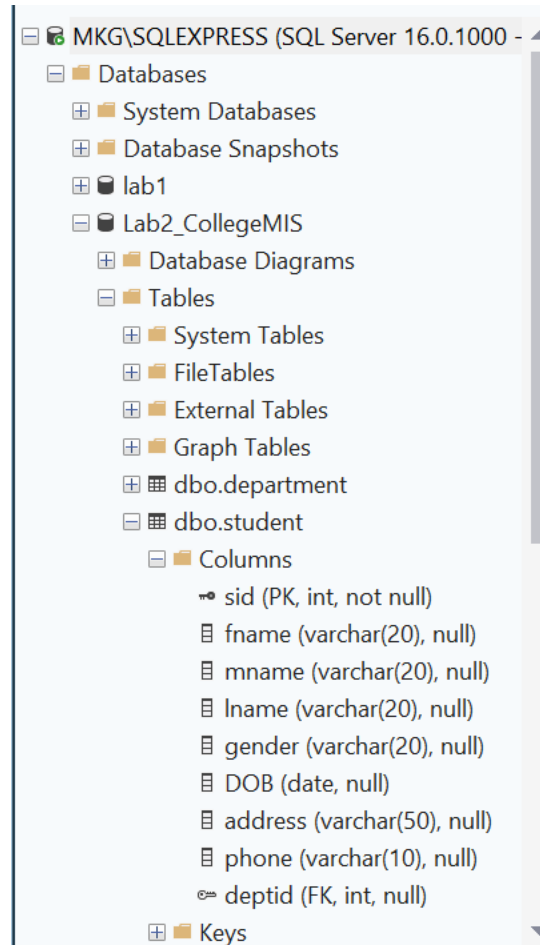


b. Student:

SQL Query:

```
CREATE TABLE student(  
sid INT PRIMARY KEY,  
fname VARCHAR(20),  
mname VARCHAR(20),  
lname VARCHAR(20),  
gender VARCHAR(20),  
DOB DATE,  
address VARCHAR(50),  
phone VARCHAR(10),  
deptid INT,  
FOREIGN KEY (deptid) REFERENCES department(deptid)  
);
```

OUTPUT:

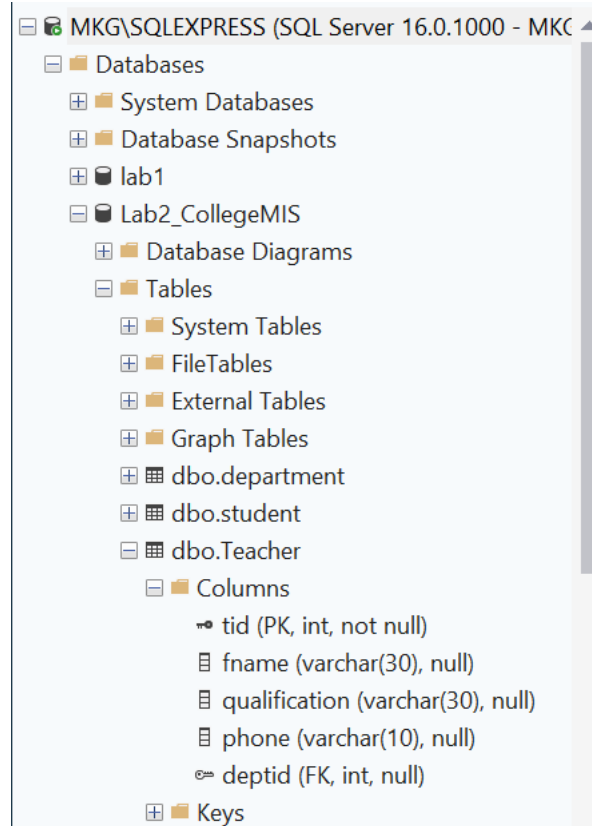


c. Teacher:

SQL Query:

```
CREATE TABLE Teacher(  
tid INT PRIMARY KEY,  
fname VARCHAR(30),  
qualification VARCHAR(30),  
phone VARCHAR(10),  
deptid INT,  
FOREIGN KEY (deptid) REFERENCES department(deptid));
```

OUTPUT:

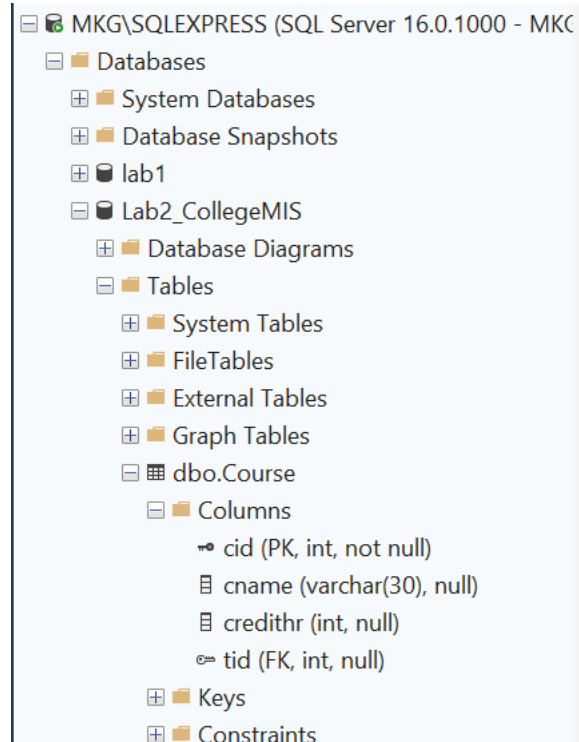


d. Course:

SQL Query:

```
CREATE TABLE Course(  
cid INT PRIMARY KEY,  
cname VARCHAR(30),  
credithr INT,  
tid INT,  
FOREIGN KEY (tid) REFERENCES Teacher(tid)  
);
```

OUTPUT:

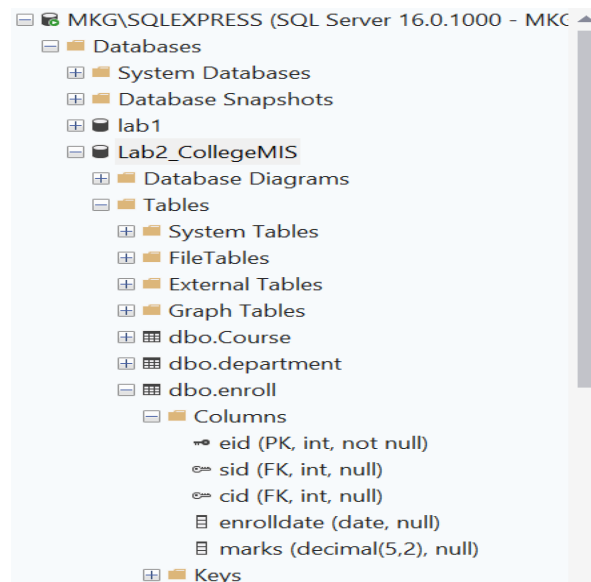


e. Enroll Table:

SQL Query:

```
CREATE TABLE enroll(  
  eid INT PRIMARY KEY,  
  sid INT,  
  FOREIGN KEY (sid) REFERENCES student(sid),  
  cid INT,  
  FOREIGN KEY (cid) REFERENCES Course(cid),  
  enrolldate DATE,  
  marks DECIMAL(5,2)  
);
```

OUTPUT:



3. Insert 10/10 records to each of the table.

a. Department:

Query:

```
INSERT INTO department VALUES
(101,'Science','Sam Acharya','R1'),
(102,'Mathematics','Sita Bhatt','R2'),
(103,'Physics','Hari Acharya','R3'),
(104,'Chemistry','Ram Bhatt','R4'),
(105,'English','Bina Gaire','R5'),
(106,'Economics','Mohan KC','R6'),
(107,'Management','Pratiksha KC','R7'),
(108,'Biology','Naman Thapa','R8'),
(109,'History','Rajiv Yadav','R9'),
(110,'Geography','Sushil Adhikari','R10');
SELECT *FROM department;
```

OUTPUT:

	deptid	deptname	HOD	room
1	101	Science	Sam Acharya	R1
2	102	Mathematics	Sita Bhatt	R2
3	103	Physics	Hari Acharya	R3
4	104	Chemistry	Ram Bhatt	R4
5	105	English	Bina Gaire	R5
6	106	Economics	Mohan KC	R6
7	107	Management	Pratiksha KC	R7
8	108	Biology	Naman Thapa	R8
9	109	History	Rajiv Yadav	R9
10	110	Geography	Sushil Adhikari	R10

b. Student:

Query:

```
INSERT INTO student VALUES
(200,'Arun','Prasad','Sapkota','Male','2002-01-15','Kathmandu','98000000001',104),
(201,'Ramila','','Rai','Female','2001-02-20','Pokhara','98000000002',108),
(202,'Ram','','Bhatt','Male','2002-03-18','Lalitpur','98000000003',101),
(203,'Hira','','Gurung','Female','2001-04-10','Bhaktapur','98000000004',103),
(204,'Aditya','','Karki','Male','2002-05-22','Chitwan','98000000005',105),
(205,'Bina','','Lamichane','Female','2001-06-12','Butwal','98000000006',101),
(206,'Nabin','','Sapkota','Male','2002-07-07','Dharan','98000000007',106),
(207,'Rita','','Joshi','Female','2001-08-14','Hetauda','98000000008',107),
(208,'Kamal','Kumar','Adhikari','Male','2002-09-30','Janakpur','98000000009',102),
(209,'Sunita','','Shrestha','Female','2001-10-25','Nepalgunj','98000000010',109);
SELECT *
FROM student;
```

OUTPUT:

Results		Messages							
	sid	fname	mname	lname	gender	DOB	address	phone	deptid
1	200	Arun	Prasad	Sapkota	Male	2002-01-15	Kathmandu	9800000001	104
2	201	Ramila		Rai	Female	2001-02-20	Pokhara	9800000002	108
3	202	Ram		Bhatt	Male	2002-03-18	Lalitpur	9800000003	101
4	203	Hira		Gurung	Female	2001-04-10	Bhaktapur	9800000004	103
5	204	Aditya		Karki	Male	2002-05-22	Chitwan	9800000005	105
6	205	Bina		Lamichane	Female	2001-06-12	Butwal	9800000006	101
7	206	Nabin		Sapkota	Male	2002-07-07	Dharan	9800000007	106
8	207	Rita		Joshi	Female	2001-08-14	Hetauda	9800000008	107
9	208	Kamal	Kumar	Adhikari	Male	2002-09-30	Janakpur	9800000009	102
10	209	Sunita		Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109

c. Teacher:

Query:

INSERT INTO Teacher VALUES

(300,'Amit Sapkota','MSc CS','9810000001',104),
(301,'Bina Gaire','MSc Math','9810000002',101),
(302,'Ramesh Thapa','MSc Physics','9810000003',107),
(303,'Susmita Karki','MSc Chemistry','9810000004',101),
(304,'Deepak Gyawali','MA English','9810000005',109),
(305,'Mina Lama','MA Economics','9810000006',102),
(306,'Bikash Joshi','MBA','9810000007',106),
(307,'Nima Poudel','MSc Biology','9810000008',103),
(308,'Kiran Adhikari','MA History','9810000009',108),
(309,'Samay Shah','MA Geography','9810000010',105);
SELECT *FROM Teacher;

OUTPUT:

Results		Messages			
	tid	fname	qualification	phone	deptid
1	300	Amit Sapkota	MSc CS	9810000001	104
2	301	Bina Gaire	MSc Math	9810000002	101
3	302	Ramesh Thapa	MSc Physics	9810000003	107
4	303	Susmita Karki	MSc Chemistry	9810000004	101
5	304	Deepak Gyawali	MA English	9810000005	109
6	305	Mina Lama	MA Economics	9810000006	102
7	306	Bikash Joshi	MBA	9810000007	106
8	307	Nima Poudel	MSc Biology	9810000008	103
9	308	Kiran Adhikari	MA History	9810000009	108
10	309	Samay Shah	MA Geography	9810000010	105

d. Course:

Query:

INSERT INTO Course VALUES

(400,'Digital Logic',3,300),
(401,'Calculus',4,301),
(402,'Mathematics',3,301),
(403,'Organic Chemistry',4,303),
(404,'English',3,304),
(405,'Microeconomics',3,305),
(406,'Web Technology',4,309),
(407,'Genetics',3,307),
(408,'History',3,308),
(409,'Physical Chemistry',3,303);
SELECT *FROM Course;

OUTPUT:

	cid	cname	credithr	tid
1	400	Digital Logic	3	300
2	401	Calculus	4	301
3	402	Mathematics	3	301
4	403	Organic Chemistry	4	303
5	404	English	3	304
6	405	Microeconomics	3	305
7	406	Web Technology	4	309
8	407	Genetics	3	307
9	408	History	3	308
10	409	Physical Chemistry	3	303

e. Enroll:

Query:

```
INSERT INTO enroll VALUES
(500,204,401,'2024-01-10',85.50),
(501,200,406,'2024-01-11',88.00),
(502,208,400,'2024-01-12',79.50),
(503,202,409,'2024-01-13',91.25),
(504,207,403,'2024-01-14',76.00),
(505,201,408,'2024-01-15',84.75),
(506,205,402,'2024-01-16',89.50),
(507,209,405,'2024-01-17',92.00),
(508,203,407,'2024-01-18',81.25),
(509,202,404,'2024-01-19',87.00);
SELECT *FROM enroll;
```

OUTPUT:

	eid	sid	cid	enrolldate	marks
1	500	204	401	2024-01-10	85.50
2	501	200	406	2024-01-11	88.00
3	502	208	400	2024-01-12	79.50
4	503	202	409	2024-01-13	91.25
5	504	207	403	2024-01-14	76.00
6	505	201	408	2024-01-15	84.75
7	506	205	402	2024-01-16	89.50
8	507	209	405	2024-01-17	92.00
9	508	203	407	2024-01-18	81.25
10	509	202	404	2024-01-19	87.00

4. Find cross join of student and department tables.

Query:

```
SELECT *
FROM student
CROSS JOIN
department;
```

OUTPUT:

Results Messages

	sid	fname	mname	lname	gender	DOB	address	phone	deptid	deptid	deptname	HOD	room
1	200	Arun	Prasad	Sapkota	Male	2002-01-15	Kathmandu	9800000001	104	101	Science	Sam Acharya	R1
2	201	Ramila		Rai	Female	2001-02-20	Pokhara	9800000002	108	101	Science	Sam Acharya	R1
3	202	Ram		Bhatt	Male	2002-03-18	Lalitpur	9800000003	101	101	Science	Sam Acharya	R1
4	203	Hira		Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	101	Science	Sam Acharya	R1
5	204	Aditya		Karki	Male	2002-05-22	Chitwan	9800000005	105	101	Science	Sam Acharya	R1
6	205	Bina		Lamichane	Female	2001-06-12	Butwal	9800000006	101	101	Science	Sam Acharya	R1
7	206	Nabin		Sapkota	Male	2002-07-07	Dharan	9800000007	106	101	Science	Sam Acharya	R1
8	207	Rita		Joshi	Female	2001-08-14	Hetauda	9800000008	107	101	Science	Sam Acharya	R1
9	208	Kamal	Kumar	Adhikari	Male	2002-09-30	Janakpur	9800000009	102	101	Science	Sam Acharya	R1
10	209	Sunita		Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	101	Science	Sam Acharya	R1
11	200	Arun	Prasad	Sapkota	Male	2002-01-15	Kathmandu	9800000001	104	102	Mathematics	Sita Bhatt	R2
12	201	Ramila		Rai	Female	2001-02-20	Pokhara	9800000002	108	102	Mathematics	Sita Bhatt	R2
13	202	Ram		Bhatt	Male	2002-03-18	Lalitpur	9800000003	101	102	Mathematics	Sita Bhatt	R2
14	203	Hira		Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	102	Mathematics	Sita Bhatt	R2
15	204	Aditya		Karki	Male	2002-05-22	Chitwan	9800000005	105	102	Mathematics	Sita Bhatt	R2
16	205	Bina		Lamichane	Female	2001-06-12	Butwal	9800000006	101	102	Mathematics	Sita Bhatt	R2
17	206	Nabin		Sapkota	Male	2002-07-07	Dharan	9800000007	106	102	Mathematics	Sita Bhatt	R2
18	207	Rita		Joshi	Female	2001-08-14	Hetauda	9800000008	107	102	Mathematics	Sita Bhatt	R2
19	208	Kamal	Kumar	Adhikari	Male	2002-09-30	Janakpur	9800000009	102	102	Mathematics	Sita Bhatt	R2
20	209	Sunita		Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	102	Mathematics	Sita Bhatt	R2
21	200	Arun	Prasad	Sapkota	Male	2002-01-15	Kathmandu	9800000001	104	103	Physics	Hari Acharya	R3
22	201	Ramila		Rai	Female	2001-02-20	Pokhara	9800000002	108	103	Physics	Hari Acharya	R3
23	202	Ram		Bhatt	Male	2002-03-18	Lalitpur	9800000003	101	103	Physics	Hari Acharya	R3
24	203	Hira		Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	103	Physics	Hari Acharya	R3
25	204	Aditya		Karki	Male	2002-05-22	Chitwan	9800000005	105	103	Physics	Hari Acharya	R3
26	205	Bina		Lamichane	Female	2001-06-12	Butwal	9800000006	101	103	Physics	Hari Acharya	R3
27	206	Nabin		Sapkota	Male	2002-07-07	Dharan	9800000007	106	103	Physics	Hari Acharya	R3
28	207	Rita		Joshi	Female	2001-08-14	Hetauda	9800000008	107	103	Physics	Hari Acharya	R3
29	208	Kamal	Kumar	Adhikari	Male	2002-09-30	Janakpur	9800000009	102	103	Physics	Hari Acharya	R3
30	209	Sunita		Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	103	Physics	Hari Acharya	R3
31	200	Arun	Prasad	Sapkota	Male	2002-01-15	Kathmandu	9800000001	104	104	Chemistry	Ram Bhatt	R4
32	201	Ramila		Rai	Female	2001-02-20	Pokhara	9800000002	108	104	Chemistry	Ram Bhatt	R4
33	202	Ram		Bhatt	Male	2002-03-18	Lalitpur	9800000003	101	104	Chemistry	Ram Bhatt	R4
34	203	Hira		Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	104	Chemistry	Ram Bhatt	R4
35	204	Aditya		Karki	Male	2002-05-22	Chitwan	9800000005	105	104	Chemistry	Ram Bhatt	R4
36	205	Bina		Lamichane	Female	2001-06-12	Butwal	9800000006	101	104	Chemistry	Ram Bhatt	R4
37	206	Nabin		Sapkota	Male	2002-07-07	Dharan	9800000007	106	104	Chemistry	Ram Bhatt	R4
38	207	Rita		Joshi	Female	2001-08-14	Hetauda	9800000008	107	104	Chemistry	Ram Bhatt	R4
39	208	Kamal	Kumar	Adhikari	Male	2002-09-30	Janakpur	9800000009	102	104	Chemistry	Ram Bhatt	R4
40	209	Sunita		Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	104	Chemistry	Ram Bhatt	R4
41	200	Arun	Prasad	Sapkota	Male	2002-01-15	Kathmandu	9800000001	104	105	English	Bina Gaire	R5
42	201	Ramila		Rai	Female	2001-02-20	Pokhara	9800000002	108	105	English	Bina Gaire	R5
43	202	Ram		Bhatt	Male	2002-03-18	Lalitpur	9800000003	101	105	English	Bina Gaire	R5
44	203	Hira		Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	105	English	Bina Gaire	R5
45	204	Aditya		Karki	Male	2002-05-22	Chitwan	9800000005	105	105	English	Bina Gaire	R5
46	205	Bina		Lamichane	Female	2001-06-12	Butwal	9800000006	101	105	English	Bina Gaire	R5
47	206	Nabin		Sapkota	Male	2002-07-07	Dharan	9800000007	106	105	English	Bina Gaire	R5
48	207	Rita		Joshi	Female	2001-08-14	Hetauda	9800000008	107	105	English	Bina Gaire	R5
49	208	Kamal	Kumar	Adhikari	Male	2002-09-30	Janakpur	9800000009	102	105	English	Bina Gaire	R5
50	209	Sunita		Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	105	English	Bina Gaire	R5

51	200	Arun	Prasad	Sapkota	Male	2002-01-15	Kathmandu	9800000001	104	106	Economics	Mohan KC	R6
52	201	Ramila		Rai	Female	2001-02-20	Pokhara	9800000002	108	106	Economics	Mohan KC	R6
53	202	Ram		Bhatt	Male	2002-03-18	Lalitpur	9800000003	101	106	Economics	Mohan KC	R6
54	203	Hira		Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	106	Economics	Mohan KC	R6
55	204	Aditya		Karki	Male	2002-05-22	Chitwan	9800000005	105	106	Economics	Mohan KC	R6
56	205	Bina		Lamichane	Female	2001-06-12	Butwal	9800000006	101	106	Economics	Mohan KC	R6
57	206	Nabin		Sapkota	Male	2002-07-07	Dharan	9800000007	106	106	Economics	Mohan KC	R6
58	207	Rita		Joshi	Female	2001-08-14	Hetauda	9800000008	107	106	Economics	Mohan KC	R6
59	208	Kamal	Kumar	Adhikari	Male	2002-09-30	Janakpur	9800000009	102	106	Economics	Mohan KC	R6
60	209	Sunita		Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	106	Economics	Mohan KC	R6
61	200	Arun	Prasad	Sapkota	Male	2002-01-15	Kathmandu	9800000001	104	107	Manageme...	Pratiksha KC	R7
62	201	Ramila		Rai	Female	2001-02-20	Pokhara	9800000002	108	107	Manageme...	Pratiksha KC	R7
63	202	Ram		Bhatt	Male	2002-03-18	Lalitpur	9800000003	101	107	Manageme...	Pratiksha KC	R7
64	203	Hira		Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	107	Manageme...	Pratiksha KC	R7
65	204	Aditya		Karki	Male	2002-05-22	Chitwan	9800000005	105	107	Manageme...	Pratiksha KC	R7
66	205	Bina		Lamichane	Female	2001-06-12	Butwal	9800000006	101	107	Manageme...	Pratiksha KC	R7
67	206	Nabin		Sapkota	Male	2002-07-07	Dharan	9800000007	106	107	Manageme...	Pratiksha KC	R7
68	207	Rita		Joshi	Female	2001-08-14	Hetauda	9800000008	107	107	Manageme...	Pratiksha KC	R7
69	208	Kamal	Kumar	Adhikari	Male	2002-09-30	Janakpur	9800000009	102	107	Manageme...	Pratiksha KC	R7
70	209	Sunita		Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	107	Manageme...	Pratiksha KC	R7
71	200	Arun	Prasad	Sapkota	Male	2002-01-15	Kathmandu	9800000001	104	108	Biology	Naman Tha...	R8
72	201	Ramila		Rai	Female	2001-02-20	Pokhara	9800000002	108	108	Biology	Naman Tha...	R8
73	202	Ram		Bhatt	Male	2002-03-18	Lalitpur	9800000003	101	108	Biology	Naman Tha...	R8
74	203	Hira		Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	108	Biology	Naman Tha...	R8
75	204	Aditya		Karki	Male	2002-05-22	Chitwan	9800000005	105	108	Biology	Naman Tha...	R8

Results Messages

	sid	fname	mname	lname	gender	DOB	address	phone	deptid	deptid	deptname	HOD	room
76	205	Bina		Lamichane	Female	2001-06-12	Butwal	9800000006	101	108	Biology	Naman Tha...	R8
77	206	Nabin		Sapkota	Male	2002-07-07	Dharan	9800000007	106	108	Biology	Naman Tha...	R8
78	207	Rita		Joshi	Female	2001-08-14	Hetauda	9800000008	107	108	Biology	Naman Tha...	R8
79	208	Kamal	Kumar	Adhikari	Male	2002-09-30	Janakpur	9800000009	102	108	Biology	Naman Tha...	R8
80	209	Sunita		Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	108	Biology	Naman Tha...	R8
81	200	Arun	Prasad	Sapkota	Male	2002-01-15	Kathmandu	9800000001	104	109	History	Rajiv Yadav	R9
82	201	Ramila		Rai	Female	2001-02-20	Pokhara	9800000002	108	109	History	Rajiv Yadav	R9
83	202	Ram		Bhatt	Male	2002-03-18	Lalitpur	9800000003	101	109	History	Rajiv Yadav	R9
84	203	Hira		Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	109	History	Rajiv Yadav	R9
85	204	Aditya		Karki	Male	2002-05-22	Chitwan	9800000005	105	109	History	Rajiv Yadav	R9
86	205	Bina		Lamichane	Female	2001-06-12	Butwal	9800000006	101	109	History	Rajiv Yadav	R9
87	206	Nabin		Sapkota	Male	2002-07-07	Dharan	9800000007	106	109	History	Rajiv Yadav	R9
88	207	Rita		Joshi	Female	2001-08-14	Hetauda	9800000008	107	109	History	Rajiv Yadav	R9
89	208	Kamal	Kumar	Adhikari	Male	2002-09-30	Janakpur	9800000009	102	109	History	Rajiv Yadav	R9
90	209	Sunita		Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	109	History	Rajiv Yadav	R9
91	200	Arun	Prasad	Sapkota	Male	2002-01-15	Kathmandu	9800000001	104	110	Geography	Sushil Adhi...	R10
92	201	Ramila		Rai	Female	2001-02-20	Pokhara	9800000002	108	110	Geography	Sushil Adhi...	R10
93	202	Ram		Bhatt	Male	2002-03-18	Lalitpur	9800000003	101	110	Geography	Sushil Adhi...	R10
94	203	Hira		Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	110	Geography	Sushil Adhi...	R10
95	204	Aditya		Karki	Male	2002-05-22	Chitwan	9800000005	105	110	Geography	Sushil Adhi...	R10
96	205	Bina		Lamichane	Female	2001-06-12	Butwal	9800000006	101	110	Geography	Sushil Adhi...	R10
97	206	Nabin		Sapkota	Male	2002-07-07	Dharan	9800000007	106	110	Geography	Sushil Adhi...	R10
98	207	Rita		Joshi	Female	2001-08-14	Hetauda	9800000008	107	110	Geography	Sushil Adhi...	R10
99	208	Kamal	Kumar	Adhikari	Male	2002-09-30	Janakpur	9800000009	102	110	Geography	Sushil Adhi...	R10
100	209	Sunita		Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	110	Geography	Sushil Adhi...	R10

5. Find Inner join of student and department tables.

Query:

SELECT

s.sid,

CONCAT(s.fname,' ',COALESCE(s.mname,'"'),' ',s.lname) AS NAME,

s.gender,s.DOB,s.address,s.phone,s.deptid,d.deptname,d.HOD,d.room

FROM student AS s

INNER JOIN

department AS d

ON d.deptid = s.deptid;

OUTPUT:

Results		Messages								
	sid	NAME	gender	DOB	address	phone	deptid	deptname	HOD	room
1	200	Arun Prasad Sapkota	Male	2002-01-15	Kathmandu	9800000001	104	Chemistry	Ram Bhatt	R4
2	201	Ramila Rai	Female	2001-02-20	Pokhara	9800000002	108	Biology	Naman Thapa	R8
3	202	Ram Bhatt	Male	2002-03-18	Lalitpur	9800000003	101	Science	Sam Acharya	R1
4	203	Hira Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	Physics	Hari Acharya	R3
5	204	Aditya Karki	Male	2002-05-22	Chitwan	9800000005	105	English	Bina Gaire	R5
6	205	Bina Lamichane	Female	2001-06-12	Butwal	9800000006	101	Science	Sam Acharya	R1
7	206	Nabin Sapkota	Male	2002-07-07	Dharan	9800000007	106	Economics	Mohan KC	R6
8	207	Rita Joshi	Female	2001-08-14	Hetauda	9800000008	107	Management	Pratiksha KC	R7
9	208	Kamal Kumar Adhikari	Male	2002-09-30	Janakpur	9800000009	102	Mathematics	Sita Bhatt	R2
10	209	Sunita Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	History	Rajiv Yadav	R9

6. Find Outer join of student and department tables.

Query:

SELECT

s.sid,

CONCAT(s.fname,' ',COALESCE(s.mname,'"'),' ',s.lname) AS NAME,

s.gender,s.DOB,s.address,s.phone,s.deptid,d.deptname,d.HOD,d.room

FROM student AS s

LEFT OUTER JOIN

department AS d

ON d.deptid = s.deptid;

OUTPUT:

Results		Messages								
	sid	NAME	gender	DOB	address	phone	deptid	deptname	HOD	room
1	200	Arun Prasad Sapkota	Male	2002-01-15	Kathmandu	9800000001	104	Chemistry	Ram Bhatt	R4
2	201	Ramila Rai	Female	2001-02-20	Pokhara	9800000002	108	Biology	Naman Thapa	R8
3	202	Ram Bhatt	Male	2002-03-18	Lalitpur	9800000003	101	Science	Sam Acharya	R1
4	203	Hira Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	Physics	Hari Acharya	R3
5	204	Aditya Karki	Male	2002-05-22	Chitwan	9800000005	105	English	Bina Gaire	R5
6	205	Bina Lamichane	Female	2001-06-12	Butwal	9800000006	101	Science	Sam Acharya	R1
7	206	Nabin Sapkota	Male	2002-07-07	Dharan	9800000007	106	Economics	Mohan KC	R6
8	207	Rita Joshi	Female	2001-08-14	Hetauda	9800000008	107	Management	Pratiksha KC	R7
9	208	Kamal Kumar Adhikari	Male	2002-09-30	Janakpur	9800000009	102	Mathematics	Sita Bhatt	R2
10	209	Sunita Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	History	Rajiv Yadav	R9

7. Display all female students.

Query:

```
SELECT
s.sid,
CONCAT(s.fname,' ',COALESCE(s.mname,''),' ',s.lname) AS NAME,
s.gender,s.DOB,s.address,s.phone,s.deptid,d.deptname,d.HOD,d.room
FROM student AS s
INNER JOIN
department AS d
ON d.deptid = s.deptid
WHERE s.gender = 'Female';
```

OUTPUT:

Results

Messages

	sid	NAME	gender	DOB	address	phone	deptid	deptname	HOD	room
1	201	Ramila Rai	Female	2001-02-20	Pokhara	9800000002	108	Biology	Naman Thapa	R8
2	203	Hira Gurung	Female	2001-04-10	Bhaktapur	9800000004	103	Physics	Hari Acharya	R3
3	205	Bina Lamichane	Female	2001-06-12	Butwal	9800000006	101	Science	Sam Acharya	R1
4	207	Rita Joshi	Female	2001-08-14	Hetauda	9800000008	107	Management	Pratiksha KC	R7
5	209	Sunita Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109	History	Rajiv Yadav	R9

8. Display student and department names.

Query:

```
SELECT
CONCAT(s.fname,' ',COALESCE(s.mname,''),' ',s.lname) AS NAME,
d.deptname
FROM student AS s
INNER JOIN
department AS d
ON d.deptid = s.deptid;
```

OUTPUT:

	Results	Messages
	NAME	deptname
1	Arun Prasad Sapkota	Chemistry
2	Ramila Rai	Biology
3	Ram Bhatt	Science
4	Hira Gurung	Physics
5	Aditya Karki	English
6	Bina Lamichane	Science
7	Nabin Sapkota	Economics
8	Rita Joshi	Management
9	Kamal Kumar Adhikari	Mathematics
10	Sunita Shrestha	History

9. Display all students with their enrolled courses.

Query:

```
SELECT
    s.sid,
    CONCAT(s.fname, ' ', COALESCE(s.mname, ''), ' ', s.lname) AS NAME,
    c.cid,
    c.cname,
    e.enrolldate,
    e.marks
FROM student s
INNER JOIN enroll e
ON s.sid = e.sid
INNER JOIN course c
ON e.cid = c.cid;
```

OUTPUT:

Results		Messages				
	sid	NAME	cid	cname	enrolldate	marks
1	204	Aditya Karki	401	Calculus	2024-01-10	85.50
2	200	Arun Prasad Sapkota	406	Web Technology	2024-01-11	88.00
3	208	Kamal Kumar Adhikari	400	Digital Logic	2024-01-12	79.50
4	202	Ram Bhatt	409	Physical Chemistry	2024-01-13	91.25
5	207	Rita Joshi	403	Organic Chemistry	2024-01-14	76.00
6	201	Ramila Rai	408	History	2024-01-15	84.75
7	205	Bina Lamichane	402	Mathematics	2024-01-16	89.50
8	209	Sunita Shrestha	405	Microeconomics	2024-01-17	92.00
9	203	Hira Gurung	407	Genetics	2024-01-18	81.25
10	202	Ram Bhatt	404	English	2024-01-19	87.00

10. Count student in each department.

Query:

```
SELECT
    d.deptid,
    d.deptname,
    COUNT(s.sid) AS total_students
FROM department d
LEFT JOIN student s
ON d.deptid = s.deptid
GROUP BY d.deptid, d.deptname;
```

OUTPUT:

Results		Messages	
	deptid	deptname	total_students
1	101	Science	2
2	102	Mathematics	1
3	103	Physics	1
4	104	Chemistry	1
5	105	English	1
6	106	Economics	1
7	107	Management	1
8	108	Biology	1
9	109	History	1
10	110	Geography	0

11. Find students scoring above 90 marks.

Query:

```
SELECT
    s.sid,
    CONCAT(s.fname,' ',COALESCE(s.mname,''),' ',s.lname) AS NAME,
    e.marks
FROM student s
INNER JOIN enroll e
ON s.sid = e.sid
WHERE e.marks > 90;
```

OUTPUT:

Results		Messages	
	sid	NAME	marks
1	202	Ram Bhatt	91.25
2	209	Sunita Shrestha	92.00

12. Find students who are not enrolled in any courses.

Query:

```
SELECT
    s.sid,
    CONCAT(s.fname,' ',COALESCE(s.mname,''),' ',s.lname) AS NAME
FROM student s
LEFT JOIN enroll e
ON s.sid = e.sid
WHERE e.sid IS NULL;
```

OUTPUT:

Results		Messages	
	sid	NAME	
1	206	Nabin Sapkota	

13. List departments with more than 6 students.

Query:

--Update of student table for this question.

```
UPDATE student
SET deptid = 100
WHERE sid IN (200, 201, 203, 204, 205, 206, 207);
```

```
SELECT
    d.deptid,
    d.deptname,
    COUNT(s.sid) AS total_students
FROM department d
INNER JOIN student s
ON d.deptid = s.deptid
GROUP BY d.deptid, d.deptname
HAVING COUNT(s.sid) > 6;
```

OUTPUT:

	Results	Messages	
	deptid	deptname	total_students
1	101	Science	8

14. Display top 3 students based on marks.

Query:

```
SELECT TOP 3
```

```
    s.sid,
```

```
    CONCAT(s.fname,' ',COALESCE(s.mname,''),' ',s.lname) AS NAME,
```

```
    e.marks
```

```
FROM student s
```

```
INNER JOIN enroll e
```

```
ON s.sid = e.sid
```

```
ORDER BY e.marks DESC;
```

OUTPUT:

	Results	Messages	
	sid	NAME	marks
1	209	Sunita Shrestha	92.00
2	202	Ram Bhatt	91.25
3	205	Bina Lamichane	89.50

15. Find average marks per course.

Query:

```
SELECT
```

```
    c.cid,
```

```
    c.cname,
```

```
    AVG(e.marks) AS average_marks
```

```
FROM Course c
```

```
INNER JOIN enroll e
```

```
ON c.cid = e.cid
```

```
GROUP BY c.cid, c.cname;
```

OUTPUT:

	Results	Messages	
	cid	cname	average_marks
1	400	Digital Logic	79.500000
2	401	Calculus	85.500000
3	402	Mathematics	89.500000
4	403	Organic Chemistry	76.000000
5	404	English	87.000000
6	405	Microeconomics	92.000000
7	406	Web Technology	88.000000
8	407	Genetics	81.250000
9	408	History	84.750000
10	409	Physical Chemistry	91.250000

16. Find students who scored above average marks.

Query:

```
SELECT
    s.sid,
    CONCAT(s.fname,' ',COALESCE(s.mname,''),' ',s.lname) AS NAME,
    e.marks
FROM student s
INNER JOIN enroll e
ON s.sid = e.sid
WHERE e.marks > (
    SELECT AVG(marks)
    FROM enroll
);
```

OUTPUT:

Results		Messages	
	sid	NAME	marks
1	204	Aditya Karki	85.50
2	200	Arun Prasad Sapkota	88.00
3	202	Ram Bhatt	91.25
4	205	Bina Lamichane	89.50
5	209	Sunita Shrestha	92.00
6	202	Ram Bhatt	87.00

17. Display department wise highest marks.

Query:

```
SELECT
    d.deptid,
    d.deptname,
    MAX(e.marks) AS highest_marks
FROM department d
INNER JOIN student s
ON d.deptid = s.deptid
INNER JOIN enroll e
ON s.sid = e.sid
GROUP BY d.deptid, d.deptname;
```

OUTPUT:

Results		Messages	
	deptid	deptname	highest_marks
1	101	Science	91.25
2	102	Mathematics	79.50
3	109	History	92.00

18. Find teacher who are teaching more than one course.

Query:

--Update in the course table

UPDATE Course

SET tid = 300

WHERE cid IN (400, 401, 402);

SELECT

t.tid,

t.fname,

COUNT(c.cid) AS total_courses

FROM Teacher t

INNER JOIN Course c

ON t.tid = c.tid

GROUP BY t.tid, t.fname

HAVING COUNT(c.cid) > 1;

OUTPUT:

	Results	Messages	
	tid	fname	total_courses
1	300	Amit Sapkota	3
2	303	Susmita Karki	2

19. Find name of all student whose department start with 'C'.

Query:

SELECT

s.sid,

CONCAT(s.fname,' ',COALESCE(s.mname,''),' ',s.lname) AS NAME,

d.deptname

FROM student s

INNER JOIN department d

ON s.deptid = d.deptid

WHERE d.deptname LIKE 'C%';

OUTPUT:

	Results	Messages	
	sid	Name	
1	200	Arun Prasad Sapkota	

20. Find join of student,teacher and department.

Query:

```
SELECT
```

```
    s.sid,
```

```
    CONCAT(s.fname,' ',COALESCE(s.mname,' '),s.lname) AS student_name,
```

```
    t.tid,
```

```
    t.fname AS teacher_name,
```

```
    d.deptid,
```

```
    d.deptname
```

```
FROM student s
```

```
INNER JOIN department d
```

```
ON s.deptid = d.deptid
```

```
INNER JOIN Teacher t
```

```
ON t.deptid = d.deptid;
```

OUTPUT:

	sid	student_name	tid	teacher_name	deptid	deptname
1	200	Arun Prasad Sapkota	301	Bina Gaire	101	Science
2	200	Arun Prasad Sapkota	303	Susmita Karki	101	Science
3	201	Ramila Rai	301	Bina Gaire	101	Science
4	201	Ramila Rai	303	Susmita Karki	101	Science
5	202	Ram Bhatt	301	Bina Gaire	101	Science
6	202	Ram Bhatt	303	Susmita Karki	101	Science
7	203	Hira Gurung	301	Bina Gaire	101	Science
8	203	Hira Gurung	303	Susmita Karki	101	Science
9	204	Aditya Karki	301	Bina Gaire	101	Science
10	204	Aditya Karki	303	Susmita Karki	101	Science
11	205	Bina Lamichane	301	Bina Gaire	101	Science
12	205	Bina Lamichane	303	Susmita Karki	101	Science
13	206	Nabin Sapkota	301	Bina Gaire	101	Science
14	206	Nabin Sapkota	303	Susmita Karki	101	Science
15	207	Rita Joshi	301	Bina Gaire	101	Science
16	207	Rita Joshi	303	Susmita Karki	101	Science
17	208	Kamal Kumar Adhik...	305	Mina Lama	102	Mathem...
18	209	Sunita Shrestha	304	Deepak Gya...	109	History

21. Change address of all student to Pokhara who enroll in computer department.

Query:

```
UPDATE student
```

```
SET address = 'Pokhara'
```

```
WHERE deptid = (
```

```
    SELECT deptid
```

```
    FROM department
```

```
    WHERE deptname = 'Computer Science'
```

```
);
```

```
SELECT *FROM student;
```

OUTPUT:

Results		Messages							
	sid	fname	mname	Iname	gender	DOB	address	phone	deptid
1	200	Arun	Prasad	Sapkota	Male	2002-01-15	Kathmandu	9800000001	101
2	201	Ramila		Rai	Female	2001-02-20	Pokhara	9800000002	101
3	202	Ram		Bhatt	Male	2002-03-18	Lalitpur	9800000003	101
4	203	Hira		Gurung	Female	2001-04-10	Bhaktapur	9800000004	101
5	204	Aditya		Karki	Male	2002-05-22	Chitwan	9800000005	101
6	205	Bina		Lamichane	Female	2001-06-12	Butwal	9800000006	101
7	206	Nabin		Sapkota	Male	2002-07-07	Dharan	9800000007	101
8	207	Rita		Joshi	Female	2001-08-14	Hetauda	9800000008	101
9	208	Kamal	Kumar	Adhikari	Male	2002-09-30	Janakpur	9800000009	102
10	209	Sunita		Shrestha	Female	2001-10-25	Nepalgunj	9800000010	109